

BMJ Open Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease: a population-based analysis of NHANES, 2007–2018

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ABSTRACT

Objective Although body mass index (BMI)-defined non-obesity metabolic dysfunction-associated steatotic liver disease (MASLD) predicts poor prognosis, the role of visceral adiposity and weight-based phenotypes is unclear. This study aimed to examine the clinical correlates of four unique MASLD phenotypes based on BMI and waist-to-height ratio (WHtR) and to assess the prognostic value of this classification with respect to mortality.

Design Population-based cross-sectional study with prospective mortality follow-up.

Setting Nationally representative US population derived from the National Health and Nutrition Examination Survey (NHANES), 2007–2018.

Participants 6300 adults with MASLD, sampled from NHANES 2007 to 2018.

Methods Participants were allocated into four phenotypes based on obesity/lean and central adiposity status. Obesity was defined as BMI ≥ 30 kg/m², while those without obesity were determined as lean. High central adiposity was defined as WHtR ≥ 0.6 .

Primary outcome measure The primary outcome was all-cause mortality. This was ascertained through National Death Index linkage with follow-up censored at 31 December 2019.

Results Among adults with MASLD, 72.1% had obesity with high central adiposity (mean follow-up: 6.9 $\pm$ 3.5 years). The non-obesity, high central adiposity phenotype (9.6%) exhibited the highest rates of hypertension (60.2%), type 2 diabetes (31.8%) and chronic kidney disease (23.3%). This phenotype portended the poorest 6-year survival ($p < 0.001$) and was independently associated with higher all-cause mortality (adjusted HR (aHR) 3.1, 95% CI 1.3 to 7.7). A J-shaped association was observed between BMI and mortality, with increasing WHtR linked to elevated risk.

Conclusion Non-obesity MASLD with high central adiposity confers the poorest survival and cardiometabolic burden.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This nationally representative, population-based study leveraged National Health and Nutrition Examination Survey (NHANES) 2007–2018 data, providing robust estimates of cardiometabolic burden and mortality risk across distinct obesity-central adiposity phenotypes in the metabolic dysfunction-associated steatotic liver disease (MASLD) population.
- ⇒ Integrating waist-to-height ratio, alongside body mass index (BMI), enabled nuanced characterisation of body fat distribution beyond traditional BMI-based measurements, enhancing the identification of high-risk non-obesity, high central adiposity MASLD phenotypes undetectable under conventional anthropometric frameworks alone.
- ⇒ BMI, waist circumference and metabolic covariates were measured at a single time point, precluding evaluation of the impact of longitudinal changes in adiposity on prognostic outcomes.
- ⇒ Case ascertainment for MASLD relied on externally validated serum-based composite scores in lieu of imaging or biopsy, introducing potential for misclassification.
- ⇒ The study population was demographically homogeneous, which may limit extrapolation to other ethnic populations with different patterns of adiposity.

INTRODUCTION

Metabolic dysfunction-associated steatotic liver disease (MASLD) poses a major public health challenge, affecting more than one-third of the adult population globally, in parallel with the obesity and type 2 diabetes (T2DM) epidemic.^{1–3} Previously recognised as non-alcoholic fatty liver disease (NAFLD),

the MASLD nomenclature change represents a shift from the exclusionary NAFLD framework to being more metabolic-centric, underpinning the pathophysiology of steatotic liver disease.³ Obesity is an antecedent risk factor of MASLD, where intrahepatic triglyceride accumulation results from metabolic disturbances, including dysregulated fatty acid metabolism, insulin resistance-mediated de novo lipogenesis and mitochondrial dysfunction.³ Ectopic fat deposition in the liver and surrounding tissues has been linked with endothelial dysfunction, systemic inflammatory response, potentiating atherogenesis and intramyocardial inflammation.^{3 4} Worsening MASLD severity has been associated with a higher incidence of cardiometabolic diseases and increasing cardiovascular disease (CVD) severity.⁵ MASLD remains an underappreciated risk factor for myocardial infarction (MI), heart failure, stroke and mortality.⁵⁻⁸

However, non-obesity MASLD has paradoxically been associated with increased rates of advanced hepatic fibrosis, all-cause mortality, as well as liver-related and cardiovascular mortality, compared with MASLD with obesity.^{9 10} However, it is important to consider that body mass index (BMI) is a heterogeneous anthropometric parameter with poor reflection of body fat distribution.¹¹ Interests in anthropometric measures of central adiposity, such as waist circumference, waist-to-height ratio (WHtR) and waist-to-hip ratio, have gained traction in recent years, with these obesity measures demonstrating better discriminating capacity than BMI in predicting cardiovascular outcomes.¹²⁻¹⁴ In particular, WHtR has demonstrated superior efficacy in reflecting visceral adiposity and predicting metabolic syndromes such as central adiposopathy, T2DM and hypertension, compared with waist circumference and BMI.¹²⁻¹⁴

Although there has been emerging interest in non-obesity MASLD phenotype in recent years,¹⁵ there is a paucity of data on the complex interplay between obesity (defined by BMI) and central adiposity (determined by WHtR) on prognostic outcomes in the general population with MASLD.⁵ This study seeks to examine the clinical correlates of the four unique MASLD phenotypes based on BMI and WHtR, as well as its prognostic mortality value and association with CVD in a population-based cohort across the past decade. The examination of the additive prognostication with the use of central adiposity markers can potentially guide risk stratification beyond the traditional BMI-defined MASLD phenotypes.

METHODS

Study population

This study used data from the National Health and Nutrition Examination Survey (NHANES), a cross-sectional survey conducted to collect health and nutrition data from the US population between 2007 and 2018. NHANES employs a stratified, multistage sampling study design to ensure representation of the general non-institutionalised population. Demographic, baseline physical parameters,

medical history and biochemical results were collected using a structured interview in the patients' homes with a subsequent standardised health examination at a mobile examination centre for physical examinations and laboratory tests. The NHANES cycles from 2007 to 2018 were selected based on the availability of all variables required for MASLD ascertainment via the Fatty Liver Index (FLI). Earlier NHANES cycles (pre-2007) did not uniformly capture all requisite biochemical and anthropometric variables necessary for MASLD ascertainment. Cycles beyond 2018 were excluded, as the National Death Index mortality linkage data, used for primary outcome ascertainment, were censored at 31 December 2019, providing insufficient follow-up duration for meaningful mortality analyses in more recent cycles. All participants provided written informed consent, and the NHANES protocol was certified and approved by the National Center for Health Statistics Ethics Review Board (ERB). The study inclusion criteria were participants aged >18 years with evidence of MASLD. Individuals with significant alcohol consumption—defined as more than two drinks per day for men and more than one drink per day for women—were excluded from the analysis (online supplemental figure 1).¹⁶ Participants were grouped into four phenotypes based on obesity/non-obesity and central adiposity status. Obesity was defined as BMI ≥ 30 kg/m² (≥ 25 kg/m² for Asians). High central adiposity was defined as a WHtR ≥ 0.6.^{17 18} Group I included participants with obesity but low central adiposity; group II included participants with obesity and high central adiposity; group III included participants with non-obesity and low central adiposity; and group IV included participants with non-obesity but high central adiposity.

Study definitions

MASLD was defined as the presence of hepatic steatosis, accompanied by at least one of the following cardiometabolic criteria (i) BMI ≥ 25 kg/m² or waist circumference > 94 cm in men and > 80 cm in women; (ii) fasting plasma glucose ≥ 5.6 mmol/L or haemoglobin A1C (HbA1c) ≥ 5.7% or the use of antidiabetic medications; (iii) average blood pressure ≥ 130/85 mm Hg or on antihypertensive treatment; (iv) plasma triglycerides ≥ 150 mg/dL (≥ 1.70 mmol/L) or lipid-lowering therapy or (v) plasma high-density lipoprotein (HDL) cholesterol ≤ 40 mg/dL (< 1.0 mmol/L) for men or < 50 mg/dL (< 1.3 mmol/L) for women or specific treatment targeting HDL cholesterol. The FLI was used to determine the presence of hepatic steatosis. The FLI equation was as follows:
$$FLI = \left(e^{0.953 \cdot \log_e(\text{triglycerides [mg/dL]}) + 0.139 \cdot \text{BMI [kg/m}^2\text{]} + 0.718 \cdot \log_e(\text{gamma-glutamyl-transferase [U/L]}) + 0.053 \cdot \text{waist circumference [cm]} - 15.745} \right) / \left(1 + e^{0.953 \cdot \log_e(\text{triglycerides [mg/dL]}) + 0.139 \cdot \text{BMI [kg/m}^2\text{]} + 0.718 \cdot \log_e(\text{gamma-glutamyl-transferase [U/L]}) + 0.053 \cdot \text{waist circumference [cm]} - 15.745} \right) \cdot 100$$
 A threshold of FLI ≥ 60 was used to determine the presence of hepatic steatosis. This has been externally validated in population studies with ultrasound-proven hepatic steatosis,^{19 20} with a sensitivity of 61%, specificity of 86%, and positive likelihood ratio of 4.3.²¹

The definitions of comorbidities in the NHANES study have been previously described.^(6, 9) The poverty-to-income ratio (PIR) was used to classify participants into socioeconomic categories: low income (PIR<1), middle income (1≤PIR≤4) and high income (PIR>4).

Study outcomes

The primary outcome was all-cause mortality, and the secondary outcome was cardiovascular mortality. Mortality status was determined through linkage with the National Death Index, with follow-up censored on 31 December 2019, and cause of death classified using International Classification of Diseases, 10th Revision (ICD-10) codes.²²

Statistical analysis

Statistical analyses were performed using R V.4.4.2 RStudio (<https://cran.r-project.org/src/base/R-4/R-4.4.2.tar.gz>). Group comparisons were conducted using Pearson's chi-squared test for categorical variables, one-way analysis of variance (ANOVA) for normally distributed continuous variables, and the Kruskal-Wallis test for non-normally distributed continuous variables. Continuous variables were presented as mean±SD for normally distributed data and median (IQR) for non-normally distributed data; categorical variables were presented as frequency (percentage). Kaplan-Meier survival curves were used to assess long-term all-cause mortality, stratified by BMI/WHtR status, and differences were evaluated using the log-rank test. HRs were calculated using Cox proportional hazards regression to assess the association between BMI/WHtR categories and all-cause mortality, adjusting for age, sex, dyslipidaemia, hypertension, T2DM, chronic kidney disease (CKD), MI, cancer and the use of anti-hypertensive, antidiabetic and lipid-lowering medications, as adjudicated in previous studies.^{23–25} Model 2 included all covariates in model 1 with additional adjustment for smoking status and sedentary activity duration. To examine the association of BMI and WHtR with long-term all-cause mortality, restricted cubic splines were constructed. Two sensitivity analyses were performed: (i) excluding participants who died within the first year of follow-up to reduce potential reverse causation and (ii) redefining hepatic steatosis using the triglyceride–glucose (TyG) index based on a previously reported cut-off.²⁶ A p value<0.05 was considered statistically significant. Participants with missing data for key variables required for exposure classification or outcome analysis were excluded from the study.

Ethical approval

NHANES 2007–2018 was approved annually by the National Center for Health Statistics (NCHS) ERB (Protocol #2005–2006 for 2007–2010, #2011-17 for 2011–2016 and #2018-01 for 2017–2018), and all participants provided written informed consent. Institutional review was not required because the analysis used de-identified, publicly accessible NHANES data and involved no participant contact or intervention.

RESULTS

Baseline characteristics

From 2007 to 2018, a total of 6300 participants with MASLD were included in the study analysis. The group II (obesity and high central adiposity) accounted for the largest proportion of the MASLD cohort (72.1%), followed by group III (non-obesity and low central adiposity; 12.2%), group IV (non-obesity but high central adiposity; 9.6%) and group I (obesity but low central adiposity; 6.1%). The mean age of the cohort was 51±17 years, and 47.2% were female. The mean follow-up duration was 6.9±3.5 years. The baseline characteristics of the study cohort are summarised in [table 1](#) and online supplemental table 1.

MASLD with non-obesity and high central adiposity phenotype

Individuals in group IV were of older age (62±15 years), followed by those in group II (51±17 years), group III (49±15 years) and group I (40±14 years, p<0.001). Individuals in group IV had the highest prevalence of comorbidities, including hypertension (60.2%, p<0.001), T2DM (31.8%, p<0.001), dyslipidaemia (67.4%, p<0.001), CKD (23.3%, p<0.001), stroke (6.3%, p<0.001) and chronic obstructive pulmonary disease (8.7%, p<0.001). They were associated with the highest mean fasting plasma glucose (p<0.001) and HbA1c (p<0.001), as well as the lowest estimated glomerular filtration rate (p<0.001) across the MASLD phenotypes. The highest proportion of individuals with low-middle income had the group IV phenotype (p<0.001).

Individuals with group IV MASLD phenotype had the highest prescription rates of anti-hypertensive medications (p<0.001), antidiabetic agents (p<0.001), lipid-lowering drugs (p<0.001) and aspirin (p<0.001). There was no significant difference in prescription rates of sodium-glucose cotransporter-2 inhibitor (SGLT2i, p=0.298) across the MASLD phenotypes ([table 1](#) and online supplemental table 1).

Study outcomes

Individuals with group IV MASLD phenotype portended the highest all-cause mortality rates (19.5%), followed by those in group III (9.6%), group II (8.5%) and group I (1.3%, p<0.001) ([table 2](#)). Kaplan-Meier curves demonstrated that individuals with group IV MASLD phenotype had the least favourable survival followed by group III, group II and group I over 6 years (p<0.001) ([figure 1](#)). In Cox regression, compared with group I, group III (adjusted HR, aHR: 3.0, 95% CI 1.2 to 7.5, p=0.017) and group IV (aHR: 2.9, 95% CI 1.2 to 7.2, p=0.022) were associated with higher all-cause mortality after full adjustment. The association for group II was attenuated and not statistically significant after adjustment (aHR: 2.3, 95% CI 0.9 to 5.6, p=0.067). The spline curve showed a J-shaped trend towards reduced all-cause mortality as BMI increased, while higher WHtR was associated with an increased risk of all-cause mortality ([figure 2](#)). Cardiovascular mortality was higher in groups II–IV compared with group I (2.5%, 2.5% and 4.3% vs 0.3%). However,

Table 1 Baseline clinical characteristics of the MASLD cohort, stratified by BMI/WHtR status

	Overall (n=6300)	Group I (BMI $\geq$ 25 kg/m ² ; WHtR<0.6) n=386	Group II (BMI $\geq$ 25 kg/m ² ; WHtR $\geq$ 0.6) n=4541	Group III (BMI<25 kg/m ² ; WHtR<0.6) n=769	Group IV (BMI<25 kg/m ² ; WHtR $\geq$ 0.6) (n=604)	P value
Baseline demographics						
Age, year	51.0 (16.6)	40.2 (13.8)	50.9 (16.7)	49.1 (14.6)	61.5 (14.7)	<0.001
Female	2973 (47.2)	60 (15.5)	2529 (55.7)	112 (14.6)	272 (45.0)	<0.001
Waist circumference, cm	111.0 (104.0–120.0)	101.0 (97.3–105.0)	115.0 (108.0–123.0)	99.1 (95.1–102.0)	105.0 (101.0–109.0)	<0.001
BMI, kg/m ²	33.2 (30.2–37.4)	30.9 (30.1–32.0)	35.2 (32.5–39.2)	27.6 (26.4–28.7)	28.8 (27.9–29.4)	<0.001
WHtR	0.7 (0.6–0.7)	0.6 (0.6–0.6)	0.7 (0.6–0.7)	0.6 (0.6–0.6)	0.6 (0.6–0.7)	<0.001
Systolic blood pressure, mm Hg	124.0 (114.0–135.0)	122.0 (115.0–131.0)	124.0 (114.0–136.0)	123.0 (115.0–133.0)	127.0 (115.0–141.0)	<0.001
Diastolic blood pressure, mm Hg	72.0 (64.7–79.3)	75.3 (67.8–82.0)	72.0 (64.0–79.3)	73.3 (66.7–79.3)	70.0 (61.3–77.3)	<0.001
Current smoking	1210 (40.6)	75 (52.4)	795 (39.0)	218 (49.0)	122 (34.4)	0.045
Family history of coronary heart disease	904 (15.0)	26 (7.2)	707 (16.3)	96 (13.0)	75 (12.8)	<0.001
Poverty-to-income ratio						<0.001
High income	1264 (23.0)	125 (17.0)	852 (23.9)	191 (20.2)	96 (23.9)	
Middle income	3141 (54.9)	158 (46.3)	2290 (55.5)	375 (52.9)	318 (58.5)	
Low income	1316 (22.1)	58 (36.7)	985 (20.6)	143 (26.9)	130 (17.6)	
Past medical history						
Hypertension	3238 (52.6)	129 (34.2)	2451 (55.3)	304 (40.3)	354 (60.2)	<0.001
T2DM	1645 (26.1)	42 (10.9)	1300 (28.6)	111 (14.4)	192 (31.8)	<0.001
Dyslipidaemia	3612 (57.7)	212 (55.1)	2543 (56.1)	453 (60.3)	404 (67.4)	<0.001
CKD	1142 (18.1)	34 (8.8)	880 (19.4)	87 (11.3)	141 (23.3)	<0.001
Stroke	269 (4.4)	6 (1.6)	207 (4.7)	18 (2.4)	38 (6.3)	<0.001
MI	518 (8.4)	12 (3.3)	379 (8.6)	45 (6.0)	82 (13.8)	<0.001
Heart failure	283 (4.6)	4 (1.1)	216 (4.9)	21 (2.8)	42 (7.0)	<0.001
COPD	152 (5.2)	1 (0.5)	119 (5.4)	9 (3.4)	23 (8.7)	<0.001
Cancer	624 (10.1)	8 (2.2)	456 (10.3)	58 (7.7)	102 (16.9)	<0.001

Categorical variables are presented as n (%) and continuous variables are presented as mean (SD) or median (IQR).
p<0.05 is taken as statistical significance.
BMI, body mass index; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; MASLD, metabolic dysfunction-associated steatotic liver disease; MI, myocardial infarction; T2DM, type 2 diabetes; WHtR, waist-to-height ratio.

after adjustment, these associations were not statistically significant (table 2).

Sensitivity analyses

Exclusion of deaths within the first year of follow-up yielded similar findings. Compared with group I, group III (aHR: 3.1, 95% CI 1.2 to 7.6, p=0.016) and group IV (aHR: 2.9, 95% CI 1.2 to 7.3, p=0.020) remained associated with higher all-cause mortality, while the estimate for group II was attenuated and not statistically significant (aHR: 2.2, 95% CI 0.9 to 5.4, p=0.079) (online supplemental table 2). Using the triglyceride–glucose index to define steatosis showed consistent results. Group III (aHR: 3.1, 95% CI 1.3 to 7.7, p=0.015) and group IV (aHR: 3.1, 95% CI 1.2 to 7.6, p=0.015) remained associated with higher all-cause mortality, whereas the estimate for group II was not statistically significant (aHR: 2.4, 95% CI 1.0 to

5.7, p=0.059) (online supplemental table 3). For cardiovascular mortality, associations were not statistically significant after adjustment in either sensitivity analysis.

DISCUSSION

Non-obesity MASLD has been an emerging area of research interest given that a substantial portion of MASLD individuals are without obesity and its distinguishing courses of disease and manifestations.^{27–29} Prior studies comparing obesity and non-obesity MASLD, with the latter defined by the traditional BMI threshold of <25 kg/m², have demonstrated inconsistent evidence of its association with cardiovascular outcomes and mortality risks.^{30 31} There remains a paucity of evidence on the use of indices that reflect visceral adiposity, in conjunction with

Table 2 Cox regression analysis of all-cause mortality in the MASLD cohort, stratified by BMI/WHtR status

	Number at risk	Number of events, n (%)	Unadjusted HR (95% CI)	P value	Model 1 hour (95% CI)	P value	Model 2 hour (95% CI)	P value
All-cause mortality								
BMI/WHtR status								
Group I – BMI $\geq$ 25 kg/m ² ; WHtR<0.6	386	5 (1.3)	Reference	—	Reference	—	Reference	—
Group II – BMI $\geq$ 25 kg/m ² ; WHtR $\geq$ 0.6	4541	388 (8.5)	6.661 (2.757 to 16.09)	<0.001	2.438 (1.002 to 5.931)	0.049	2.299 (0.944 to 5.599)	0.067
Group III – BMI<25 kg/m ² ; WHtR<0.6	769	74 (9.6)	6.252 (2.527 to 15.47)	<0.001	3.175 (1.279 to 7.882)	0.013	3.029 (1.219 to 7.526)	0.017
Group IV – BMI<25 kg/m ² ; WHtR $\geq$ 0.6	604	118 (19.5)	15.13 (6.181 to 37.02)	<0.001	3.155 (1.276 to 7.805)	0.013	2.893 (1.167 to 7.169)	0.022
Cardiovascular mortality								
BMI/WHtR status								
Group I – BMI $\geq$ 25 kg/m ² ; WHtR<0.6	386	1 (0.3)	Reference	—	Reference	—	Reference	—
Group II – BMI $\geq$ 25 kg/m ² ; WHtR $\geq$ 0.6	4541	112 (2.5)	9.642 (1.346 to 69.05)	0.024	3.440 (0.475 to 24.93)	0.222	3.367 (0.464 to 24.44)	0.23
Group III – BMI<25 kg/m ² ; WHtR<0.6	769	19 (2.5)	8.113 (1.086 to 60.61)	0.041	4.157 (0.552 to 31.31)	0.167	4.083 (0.542 to 30.77)	0.172
Group IV – BMI<25 kg/m ² ; WHtR $\geq$ 0.6	604	26 (4.3)	16.75 (2.273 to 123.4)	0.006	3.376 (0.449 to 25.38)	0.237	3.256 (0.432 to 24.53)	0.252
Model 1: adjusted for age, sex, dyslipidaemia, hypertension, type 2 diabetes, chronic kidney disease, myocardial infarction and cancer and use of antihypertensive medications, glucagon-like peptide-1 receptor agonists, insulin and lipid-lowering medications. Model 2: adjusted for all model 1 covariates, with additional adjustment for smoking status and sedentary activity duration. p<0.05 is taken as statistical significance. BMI, body mass index; MASLD, metabolic dysfunction-associated steatotic liver disease; WHtR, waist-to-height ratio.								

BMI, in enhancing the risk stratification of patients with MASLD.¹⁵ Thus, this study extends the current literature by combining WHtR, an index of central adiposity, and BMI in distinguishing four unique MASLD phenotypes and their impact on prognostic outcomes in a population-based study. There are several pertinent study findings. First, MASLD with obesity and high central adiposity was the most common phenotype in the general population. However, approximately 10% of the MASLD cohort had the non-obesity but high central adiposity phenotype. Specific vulnerable subpopulations, such as those of older age, middle-to-low-income groups and increased metabolic complications, were most susceptible to the non-obesity but high central adiposity MASLD phenotype. Second, BMI and central adiposity displayed directionally opposite relationships with all-cause mortality in which individuals with lower BMI but elevated WHtR were associated with worse outcomes (figure 3).

One-fifth of the MASLD population had non-obesity phenotypes, with 43% of them exhibiting high central adiposity. While previous studies on MASLD and obesity predominantly relied on a single anthropometric measure of obesity^{15 32} or examined these obesity biomarkers in isolation from one another, this study emphasises the limitations of BMI, as the reliance on BMI alone fails to distinguish the components of body composition and body fat distribution (central or peripheral adipose depots).³³ Studies that solely used BMI had consistently reported an obesity paradox, with patients with MASLD with obesity portending a better prognosis than their counterparts with a BMI below the obesity threshold, not just in the MASLD population but also in other cardiovascular cohorts.^{34–36} Our findings have challenged this concept, highlighting that the combined measure of BMI and WHtR stratified patients with MASLD into

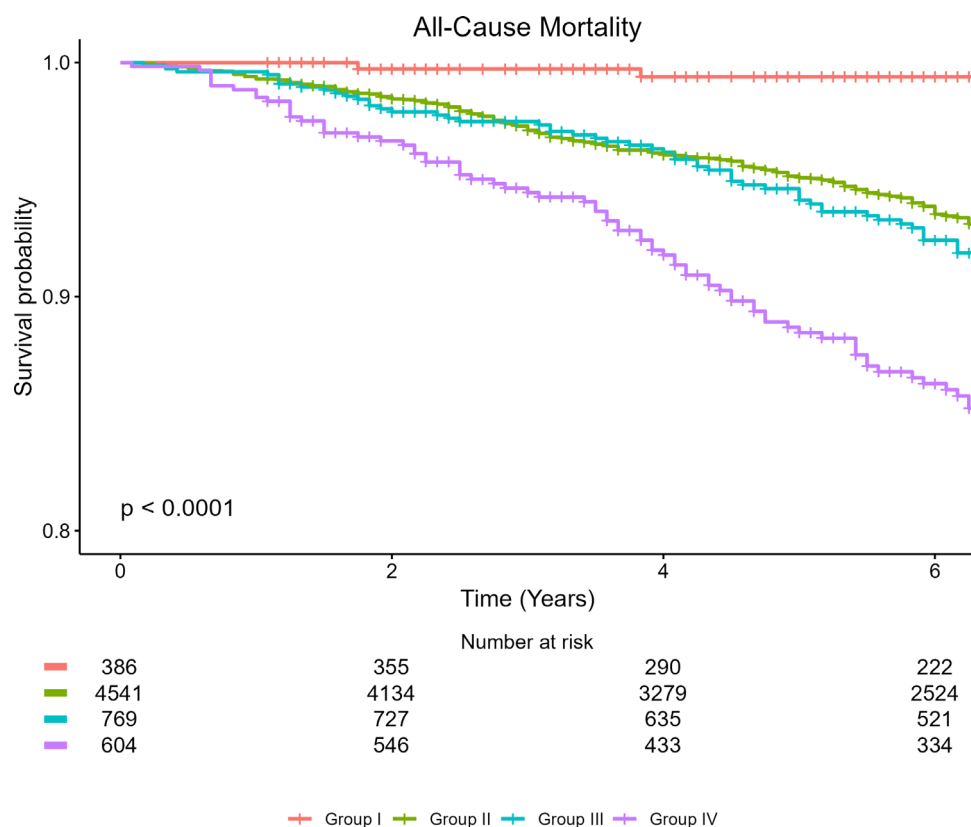


Figure 1 Kaplan-Meier curve for all-cause mortality in the MASLD cohort, stratified by BMI/WHtR status. BMI, body mass index; MASLD, metabolic dysfunction-associated steatotic liver disease; WHtR, waist-to-height ratio.

unique phenotypes with differential cardiometabolic implications.

MASLD individuals with obesity and high central adiposity tended to be older, predominantly female, with a greater burden of comorbidities including metabolic diseases, CVD and cancer, compared with MASLD counterparts with obesity but low central adiposity.^{37 38} Notably, this cardiometabolic burden was more pronounced in the MASLD with non-obesity but high central adiposity phenotypes. Our findings highlight a particularly susceptible subset of individuals with non-obesity MASLD but high central adiposity, characterised by older age, with middle-to-low income and burdened with cardiometabolic

complications. The patho-mechanistic pathway in the non-obesity MASLD phenotype with high central adiposity is poorly understood, though this is likely underpinned by impaired metabolic flexibility and maladaptive responses to increased central adiposity,³⁹ where metabolically active visceral fat depots lead to systemic chronic inflammation, increased sympathetic nervous system and renin-angiotensin-aldosterone system activities, increased hepatic lipotoxicity and direct portal delivery of free fatty acids, collectively intensifying insulin resistance.^{39–41} These interconnected mechanisms establish a deleterious metabolic milieu, resulting in vascular impairment and promoting tumourigenesis over time.^{39 42} The

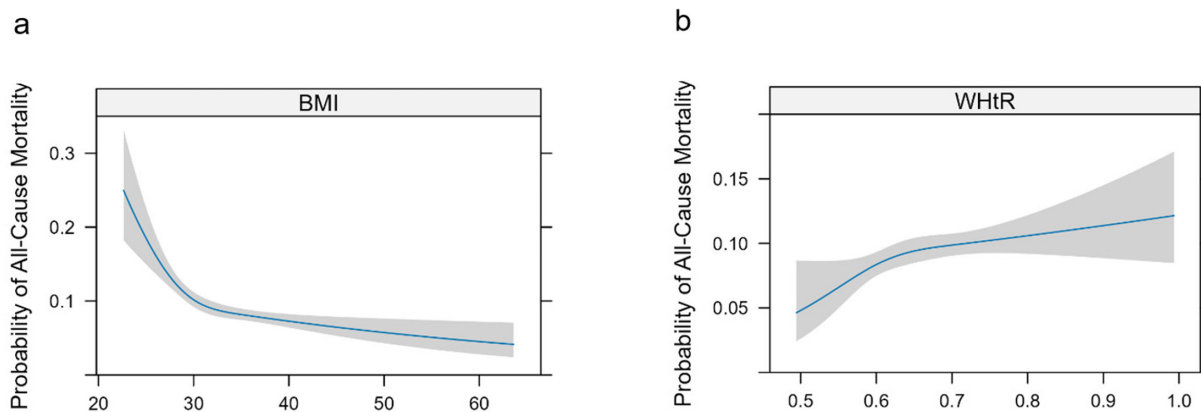


Figure 2 Cubic spline plots on the association of BMI and WHtR with the probability of all-cause mortality. BMI, body mass index; WHtR, waist-to-height ratio.

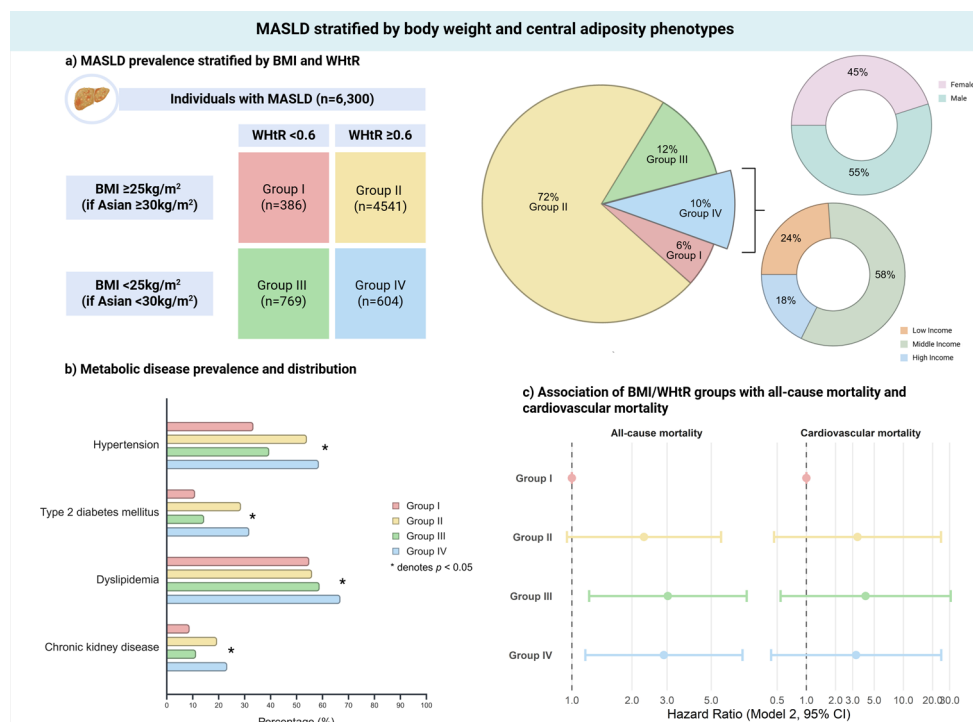


Figure 3 Central illustration—the clinical associates and prognostic outcomes of the MASLD cohort, stratified by BMI/WHtR status. BMI, body mass index; MASLD, metabolic dysfunction-associated steatotic liver disease; WHtR, waist-to-height ratio.

subcutaneous adipose tissue capacity for lipid storage in this cohort can also lead to the amplification of metabolic dysfunction by redirecting excess lipids to ectopic sites, including the liver and skeletal muscles.^{39 43} Clinicians should remain cognisant that individuals with non-obesity MASLD are at elevated risk of cardiometabolic diseases, especially in individuals with high central adiposity.^{5 44}

A striking study finding was the opposing directional trends in prognostic outcomes between BMI and WHtR, with higher BMI portending better survival, while higher WHtR associates poorer outcomes in the MASLD population.³⁶ The J-shaped association between BMI and all-cause mortality observed in our study is underpinned by the mechanistically complex, stage-dependent relationship between adiposity and MASLD progression. In early-stage MASLD, elevated BMI potentiates cardiometabolic risk via expansion of dysfunctional adipose tissue, leading to increases in pro-inflammatory cytokine release, insulin resistance and hepatic free fatty acid overload.⁴⁵ Collectively, this drives atherogenesis and systemic vascular inflammation.⁴⁵ However, as MASLD progresses towards cirrhosis, this relationship often attenuates or reverses—a phenomenon widely described as the obesity paradox.^{36 46} In advanced MASLD disease, progressive sarcopenia, unintentional weight loss and impaired hepatic protein synthesis can lead to a paradoxical state whereby lower BMI associates with increased metabolic stress, cardiovascular fragility and mortality risk.^{47 48} Moreover, individuals with advanced CVD and cancer are at higher risk of cardiac cachexia and sarcopenia,⁴⁹ and may consequently have reduced hepatic fat

accumulation and attenuated markers of inflammation such as ALT.^{47 48} Conversely, elevated BMI may confer a degree of nutritional reserve that attenuates short-term mortality risk. This complex mechanistic duality renders the relationship between conventional adiposity markers and mortality risk inherently non-linear across the MASLD disease spectrum, which, in part, explains the apparent protective effect of higher BMI observed in our survival analyses. Critically, this reinforces that BMI alone is an insufficient prognostic marker of MASLD, and the integration of central adiposity markers is important to more accurately identify cardiometabolic risk conferred by excess visceral adiposity, independent of total body weight.⁵

To further the cardiovascular-liver-metabolic health agenda, stakeholders must consider that the risk stratification of MASLD goes beyond a single anthropometric measure in isolation.⁵⁰ With emerging insights into the prognostic value of the non-obesity MASLD phenotype with high central adiposity, more accurate imaging-based biomarkers of central adiposity should be considered when risk-stratifying individuals with MASLD.^{51 52} The under-recognition of this phenotype can lead to assumptions of ‘metabolically healthy’ status, leading to missed opportunities for cardiometabolic prevention and prompt referral.⁵ More attention is needed to expand the clinical trial design to ensure the inclusion of this subgroup when evaluating treatment efficacy in MASLD. In the era of new and scalable treatments of MASLD,^{53–55} there is vast potential in emerging metabolic pharmacotherapies that target the shared pathways of visceral

adiposopathy and metabolic inflexibility,^{56–58} especially in the MASLD subpopulation with the non-obesity but high central adiposity phenotype, which may include SGLT2i and glucagon-like peptide-1 receptor agonists.^{53 59 60} As insights into MASLD continue to evolve, future focus is shifting toward the recognition of MASLD subclassifications, some of which are associated with higher CVD risk.^{61 62} While not perfect, the combination of BMI and WHtR in clinical practice may offer pragmatic and effective surrogate markers of body fat distribution,⁵ with the potential to improve risk stratification of the MASLD cohort with greater metabolic burden and cardiovascular risk.^{46 63 64}

Limitations

This study has limitations. First, all measurements included, such as waist circumference and BMI, were recorded as a single-time input, which may not fully account for the dynamic changes of anthropometric variables over time. Given the nature of the study, the impact of weight gain or loss on prognostic outcomes in MASLD could not be measured. Second, longitudinal cardiovascular outcomes were not available. Future large-scale studies are warranted to perform time-to-event analysis for cardiovascular endpoints across the MASLD phenotypes. Third, the study was based on mainly white population, thus affecting the generalisability of results to other population groups. Further evaluation in an Asian cohort is needed, especially with the remarkable variability in body fat distribution for a given BMI in Asians and the higher prevalence of non-obesity MASLD reported in Asians compared with Western populations.⁶⁵ Fourth, simple and readily available anthropometric measures, such as WHtR, were used to represent visceral adiposity. Future studies are warranted to validate these findings against more accurate imaging (eg, CT and MRI) and non-imaging-based (eg, bioelectrical impedance analysis and skinfold thickness) measures of visceral adipose tissue, which were not available in the study. The gold-standard liver biopsy or imaging was not available for the diagnosis of MASLD. Instead, externally validated serum-based composite scores were used, which enable the pragmatic alternative for larger-scale screening of MASLD in the large population-based study. Finally, vibration-controlled transient elastography data were not included in our study due to the limited follow-up duration for mortality, restricted to 3 years.

CONCLUSION

More than two-fifths of the general population with a BMI-defined non-obesity MASLD phenotype had high central adiposity. This unique MASLD phenotype portends an increased risk of cardiometabolic comorbidity, with the least favourable survival compared with other MASLD phenotypes. There is a need to transcend beyond the traditional BMI-based paradigm of categorising MASLD phenotypes by integrating readily available biomarkers of

central adiposity, such as WHtR, for more precise prognostication that can guide cardiovascular risk stratification and timely management.

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Competing interests CWleR: CWleR reports grants from the Irish Research Council, Science Foundation Ireland, Anabio, and the Health Research Board. He serves on advisory boards of Novo Nordisk, Herbalife, Gl Dynamics, Eli Lilly, Johnson; Johnson, Glia and Boehringer Ingelheim. CWleR is a member of the Irish Society for Nutrition and Metabolism outside the area of work commented on here. He was the chief medical officer and director of the Medical Device Division of Keyron in 2011. Both of these are unremunerated positions. CWleR was a previous investor in Keyron, which develops endoscopically implantable medical devices intended to mimic the surgical procedures of sleeve gastrectomy and gastric bypass. The product has only been tested in rodents, and none of Keyron's products are currently licensed. They do not have any contracts with other companies to put their products into clinical practice. No patients have been included in any of Keyron's studies, and they are not listed on the stock market. CWleR was gifted stock holdings in September 2021 and divested all stock holdings in Keyron in September, 2021. He continues to provide scientific advice to Keyron for no remuneration. AM: AM has received institutional grant support from Amgen and Novartis. MSK: MSK reports fees from Bayer and Novartis. NWSC: NWSC has received research grant support from the NUHS Seed Fund (NUHSR0/2022/R05+6/Seed-Mar/03), the National Medical Research Council Research Training Fellowship (MH 095:003/008-303), the National University of Singapore Yong Loo Lin School of Medicine's Academic Fellowship Scheme, the NUHS Clinician Scientist Program (NCSP2.0/2024/NUHS/NCWS) and the National Medical Research Council Transition Award (TA24jul-0008). All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. AM: AM has received institutional grant support from Amgen and Novartis. MSK: MSK reports fees from Bayer and Novartis. NWSC: NWSC has received research grant support from the NUHS Seed Fund (NUHSR0/2022/R05+6/Seed-Mar/03), the National Medical Research Council Research Training Fellowship (MH 095:003/008-303), the National University of Singapore Yong Loo Lin School of Medicine's Academic Fellowship Scheme, the NUHS Clinician Scientist Program (NCSP2.0/2024/NUHS/NCWS) and the National Medical Research Council Transition Award (TA24jul-0008). All other authors have no completing/competing interest to declare.

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